Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Physical AI-Driven Adaptive Prehabilitation System for Personalized Athletic Performance Optimization with Wearable Exosuit Integration**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio • Medical		*Mechanical • Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Functional Restoration / Assistive & Welfare Devices	*Rehabilitation Training Devices*	*Robotics / Automation Machinerys*	*Robot Control and Intelligence Technologies*
Concept of Proposed Technology	**Core Concept** This project develops a Physical AI-powered Prehabilitation (ML-Prehab) system that shifts training from reactive injury treatment to proactive injury prevention. By integrating advanced machine learning algorithms from research institute with a Korean SME partner's soft wearable robot platform, the system creates a closed-loop framework for optimizing human performance while preventing overuse injuries. **Key Functions** 1) Multimodal sensor fusion combining biomechanical and physiological signals 2) Reinforcement learning algorithms with safety guarantees that adapt to each user's evolving state 3) Real-time intervention delivery through an integrated soft wearable robot for adaptive			

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	mechanical assistance.
Rationale for Joint R&D Project	**Why Global Collaboration?** Research institute brings state-of-the-art optimization and federated learning algorithms, while the Korean SME partner provides a clinically validated wearable robotics platform. The collaboration combines mathematical rigor with real-world validation capability through clinical partnerships and testing infrastructure. Neither party could achieve this integration alone.
Key Words	1. Machine Learning Prehabilitation 2. Wearable Exosuit Robotics 3. Online Reinforcement Learning 4. Injury Prevention 5. Sensor Fusion

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	**R&D Objectives** 1) Adaptive Online Learning Framework ● Develop RL algorithms that handle human non-stationarity with safety guarantees, adapting to each user's physiological evolution without catastrophic forgetting. 2) Multimodal Sensor Fusion Engine ● Create a unified signal processing pipeline integrating biomechanical (sEMG, IMU, motion capture), physiological (HRV, sweat biomarkers), and subjective data streams. 3) Constrained Optimization for Injury-Risk Management ● Design optimization algorithms balancing performance maximization against injury risk constraints using Acute-to-Chronic Workload Ratio (ACWR) monitoring. 4) Wearable Robot Integration Module ● Develop bidirectional API connecting Physical AI algorithms with the wearable robot platform for real-time training modification. 5) Federated Learning for User Privacy ● Implement privacy-preserving learning across distributed user data, enabling model improvement without centralizing sensitive health information.			
R&D Target Specifications	● Injury Prediction Accuracy: ≥85% ● Algorithm Adaptation Time: Real-time personalization ● Sensor Modalities Integrated: Multi-modal fusion ● Wearable Robot Response Latency: Real-time control ● Injury Reduction: Tangible reduction in injury risk metrics of at least 10% against control group			
R&D Plan	**Phase 1: Preliminary Research (6 months)** 1) Month 1-2: Algorithm design: online RL architecture with safety constraints 2) Month 3-4: Baseline data collection (cohort of n≥50 users with multimodal sensors) 3) Month 5-6: Initial algorithm training and simulation testing **Phase 2: Main Research** 1) Year 1: Sensor fusion engine development, Physical AI algorithm deployment, pilot study 2) Year 2: Federated learning implementation, large-scale validation of Physical AI models, comparative analysis, patent applications 3) Year 3: Regulatory documentation, product integration roadmap, tech transfer &			

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	commercialization planning

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	● Lead the design and development of adaptive online RL algorithms with safety guarantees ● Develop federated learning framework for privacy-preserving model training ● Create multimodal sensor fusion algorithms and signal processing pipeline ● Design constrained optimization algorithms for injury-risk management ● Conduct theoretical analysis and convergence guarantees ● Publish research findings in top-tier venues ● Provide graduate student researchers and computational resources
Expected R&R of Korean SME Partner	● Provide access to clinically validated soft wearable robot platform ● Supply sensor hardware and integration interfaces ● Develop bidirectional API for real-time robot control ● Conduct clinical validation studies with partner rehabilitation facilities ● Lead regulatory documentation and certification processes ● Handle commercialization, manufacturing, and distribution ● Provide testing infrastructure and clinical partnerships

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Achievements** 1) First integrated ML-wearable robot platform for proactive athlete injury prevention 2) Novel online RL algorithms with convergence guarantees for human non-stationarity 3) Open-source sensor fusion toolkit for sports science research **Intellectual Property** 1) Patent applications covering algorithm-wearable robot integration, sensor fusion methods, and federated athlete modeling 2) Publications in top-tier journals **Commercialization Potential** 1) Target Market: Rehabilitation clinics, Fitness centers, Industrial safety 2) Competitive Advantage: Only solution combining safety guarantees + clinically validated wearable robot **Economic Impact** 1) New product line for the Korean SME partner with estimated 20% revenue growth 2) Reduction in injury-related costs 3) Creation of new R&D positions

6. References

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(3) Xu, J., Huang, K., Zhao, M., & Liu, J. (2024). Human-in-the-Loop Modeling and Bilateral Skill Transfer Control of Soft Exoskeleton. Sensors, 24(23), 7845.

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